

ASSIGNMENT-15.2

Backend API Development: Creating RESTful services with AI

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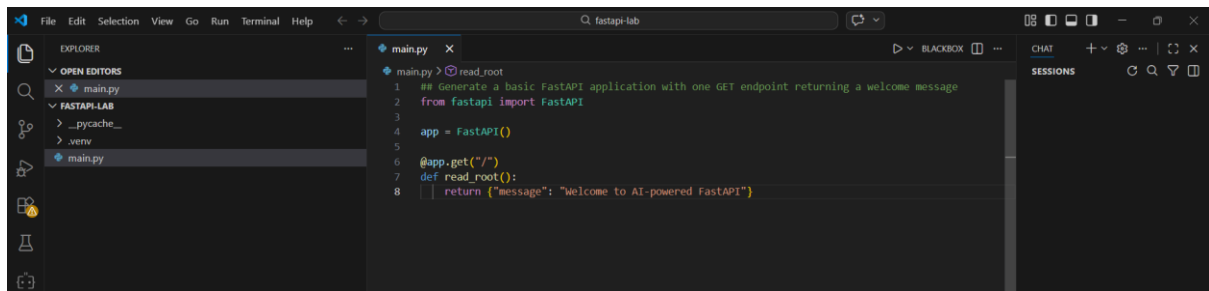
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B-52

Task Descriptor-1: (Setup Fast API Backend)

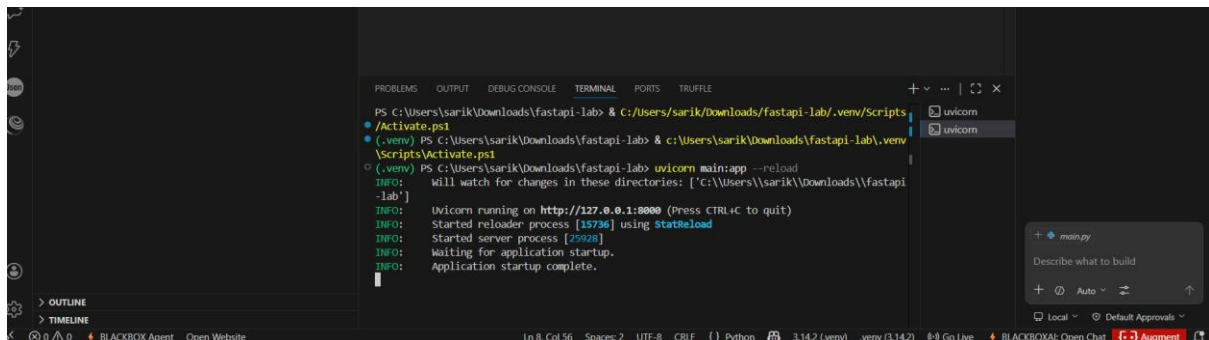
PROMPT: Generate a basic FastAPI application with one GET endpoint returning a welcome message

INPUT:

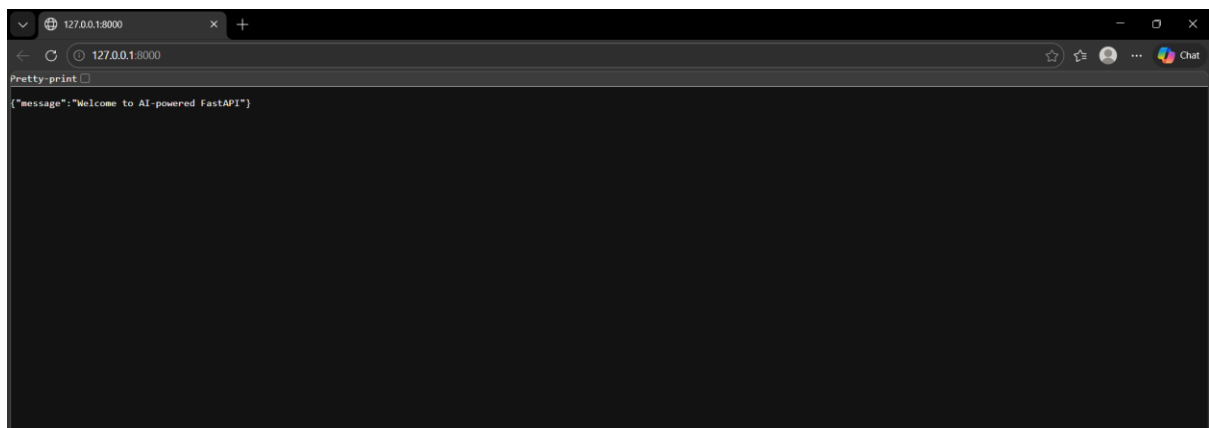


```
main.py > read_root
1  ## Generate a basic FastAPI application with one GET endpoint returning a welcome message
2  from fastapi import FastAPI
3
4  app = FastAPI()
5
6  @app.get("/")
7  def read_root():
8      return {"message": "Welcome to AI-powered FastAPI"}
```

OUTPUT:



```
PS C:\Users\sarik\Downloads\fastapi-lab> & c:\Users\sarik\Downloads\fastapi-lab\.venv\Scripts
/activate.ps1
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> & c:\Users\sarik\Downloads\fastapi-lab\.venv
Scripts/activate.ps1
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn main:app --reload
INFO: will watch for changes in these directories: ['c:\Users\sarik\Downloads\fastapi
-lab']
INFO: uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reload process [15736] using StatReload
INFO: Started server process [25928]
INFO: Waiting for application startup.
INFO: Application startup complete.
```



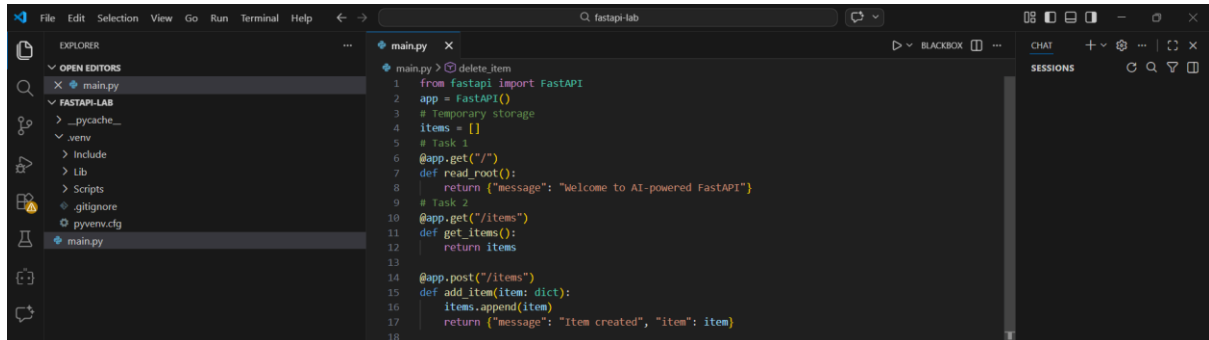
```
127.0.0.1:8000
{"message": "Welcome to AI-powered FastAPI"}
```

Explanation: FastAPI uses decorators to define API routes easily. The root endpoint returns a JSON response when accessed.

Task Descriptor- 2: (API – Create and Read Items)

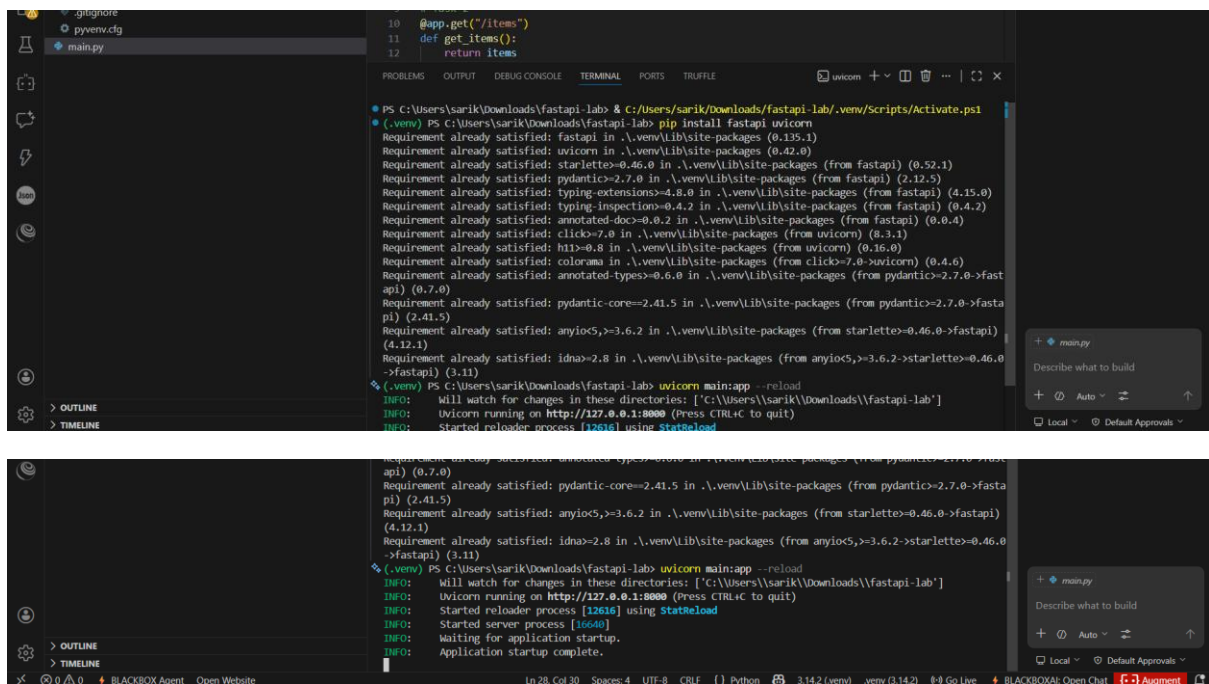
PROMPT: Create FastAPI endpoints to add and retrieve items using a Python list

INPUT:



```
1 from fastapi import FastAPI
2 app = FastAPI()
3 # Temporary storage
4 items = []
5 # Task 1
6 @app.get("/")
7 def read_root():
8     return {"message": "Welcome to AI-powered FastAPI"}
9 # Task 2
10 @app.get("/items")
11 def get_items():
12     return items
13
14 @app.post("/items")
15 def add_item(item: dict):
16     items.append(item)
17     return {"message": "Item created", "item": item}
```

OUTPUT:



```
PS C:\Users\sarik\Downloads\fastapi-lab> pip install fastapi uvicorn
Requirement already satisfied: fastapi in .\venv\Lib\site-packages (0.135.1)
Requirement already satisfied: uvicorn in .\venv\Lib\site-packages (0.42.0)
Requirement already satisfied: starlette>=0.46.0 in .\venv\Lib\site-packages (from fastapi) (0.52.1)
Requirement already satisfied: pydantic>=2.7.0 in .\venv\Lib\site-packages (from fastapi) (2.12.5)
Requirement already satisfied: typing-extensions>=4.8.0 in .\venv\Lib\site-packages (from fastapi) (4.15.0)
Requirement already satisfied: typing-inspection>=0.4.2 in .\venv\Lib\site-packages (from fastapi) (0.4.2)
Requirement already satisfied: annotated-doc>=0.0.2 in .\venv\Lib\site-packages (from fastapi) (0.0.4)
Requirement already satisfied: click>=7.0 in .\venv\Lib\site-packages (from uvicorn) (8.3.1)
Requirement already satisfied: h11>=0.8 in .\venv\Lib\site-packages (from uvicorn) (0.16.0)
Requirement already satisfied: colorama in .\venv\Lib\site-packages (from click>=7.0->uvicorn) (0.4.6)
Requirement already satisfied: annotated-types>=0.6.0 in .\venv\Lib\site-packages (from pydantic>=2.7.0->fastapi) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.5 in .\venv\Lib\site-packages (from pydantic>=2.7.0->fastapi) (2.41.5)
Requirement already satisfied: anyio<5,>=3.6.2 in .\venv\Lib\site-packages (from starlette>=0.46.0->fastapi) (4.12.1)
Requirement already satisfied: idna>=2.8 in .\venv\Lib\site-packages (from anyio<5,>=3.6.2->starlette>=0.46.0->fastapi) (3.11)
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn mainapp --reload
INFO: Will watch for changes in these directories: ['C:\Users\sarik\Downloads\fastapi-lab']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reload process [12616] using StatReload
INFO: Started server process [16040]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

POST

/items Add item

Cancel

Reset

Parameters

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "name": "Pen",
  "price": 20
}
```

Execute

Responses

Code	Description	Links
------	-------------	-------

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Pen",
    "price": 20
  }'
```

Request URL

http://127.0.0.1:8000/items

Server response

Code	Details
200	Response body <pre>{ "message": "Item created", "item": { "name": "Pen", "price": 20 } }</pre> Response headers <pre>content-length: 59 content-type: application/json date: Tue, 17 Mar 2020 15:27:29 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

GET

/items Get items

Cancel

Parameters

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/items

Server response

Code	Details
200	Response body <pre>[{ "name": "Pen", "price": 20 }, { "name": "Pen", "price": 20 }]</pre>

Responses

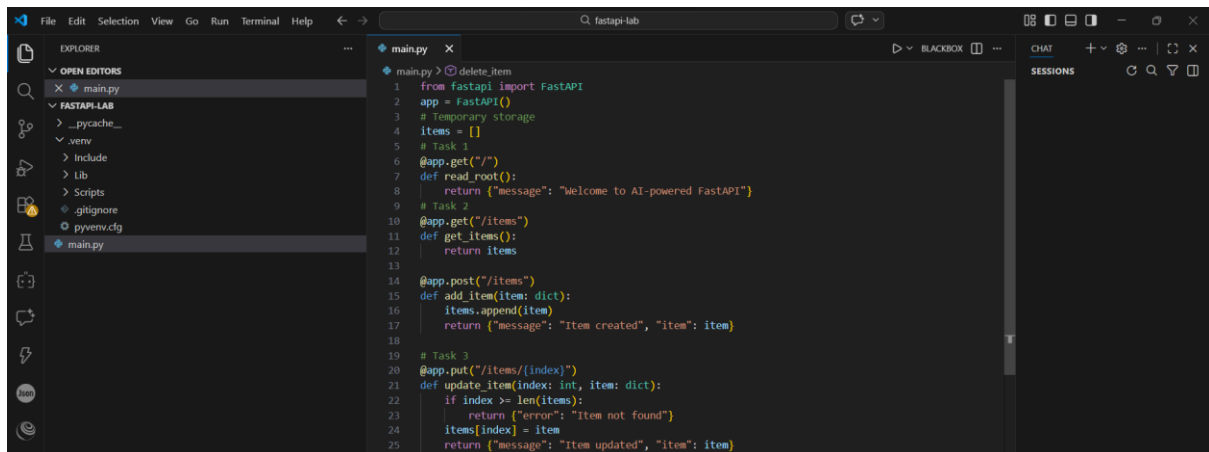
Code	Description	Links
------	-------------	-------

Explanation: GET retrieves all items stored in the list. POST adds a new item into the list.

Task Descriptor- 3: (Update Existing Item)

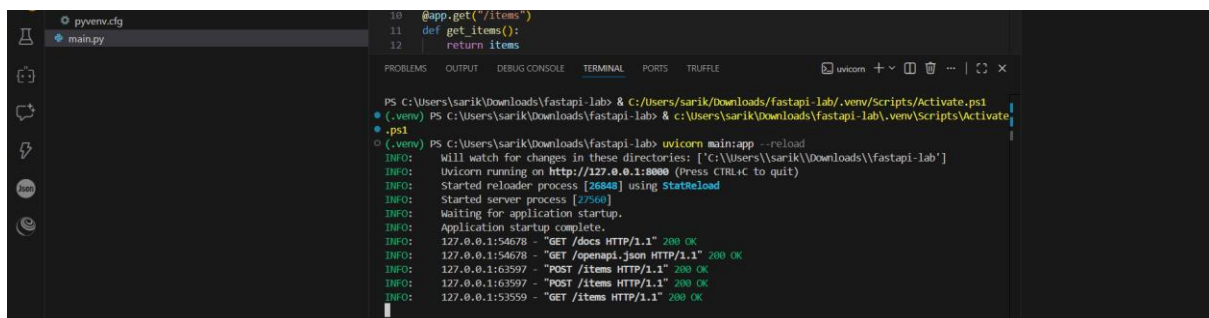
PROMPT: Create a PUT endpoint to update an item using its index.

INPUT:

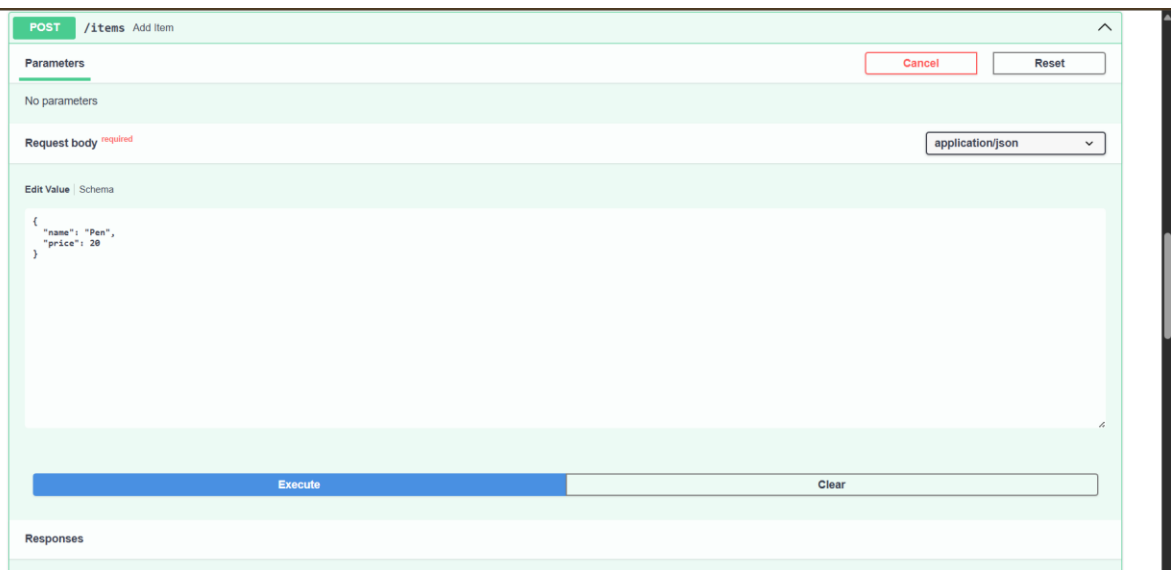


```
1 from fastapi import FastAPI
2 app = FastAPI()
3 # Temporary storage
4 items = []
5 # Task 1
6 @app.get("/")
7 def read_root():
8     return {"message": "Welcome to AI-powered FastAPI"}
9 # Task 2
10 @app.get("/items")
11 def get_items():
12     return items
13
14 @app.post("/items")
15 def add_item(item: dict):
16     items.append(item)
17     return {"message": "Item created", "item": item}
18
19 # Task 3
20 @app.put("/items/{index}")
21 def update_item(index: int, item: dict):
22     if index >= len(items):
23         return {"error": "Item not found"}
24     items[index] = item
25     return {"message": "Item updated", "item": item}
```

OUTPUT:



```
PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\Users\sarik\Downloads\fastapi-lab']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reload process [26648] using StarReload
INFO: Started server process [27560]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:54678 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:54678 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 127.0.0.1:63597 - "POST /items HTTP/1.1" 200 OK
INFO: 127.0.0.1:63597 - "POST /items HTTP/1.1" 200 OK
INFO: 127.0.0.1:53559 - "GET /items HTTP/1.1" 200 OK
```



POST /items Add Item

Parameters Cancel Reset

No parameters

Request body required application/json

Edit Value | Schema

```
{
  "name": "Pen",
  "price": 20
}
```

Execute Clear

Responses

GET /items Get Items

Parameters

Cancel

No parameters

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/items' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

CodeDetails

200

Response body

```
{
  "name": "Pen",
  "price": 20
},
{
  "name": "Pen",
  "price": 20
}
}
```

Download

PUT /items/{index} Update Item

Parameters

CancelReset

NameDescription

index * required

Integer(path)0

Request body required

application/json

Edit ValueSchema

```
{
  "name": "Pencil",
  "price": 10
}
```

Execute

ExecuteClear

Responses

Curl

```
curl -X 'PUT' \
'http://127.0.0.1:8000/items/0' \
-H 'accept: application/json' \
-H 'Content-Type: application/json' \
-d '{
  "name": "Pencil",
  "price": 10
}'
```

Request URL

```
http://127.0.0.1:8000/items/0
```

Server response

CodeDetails

200

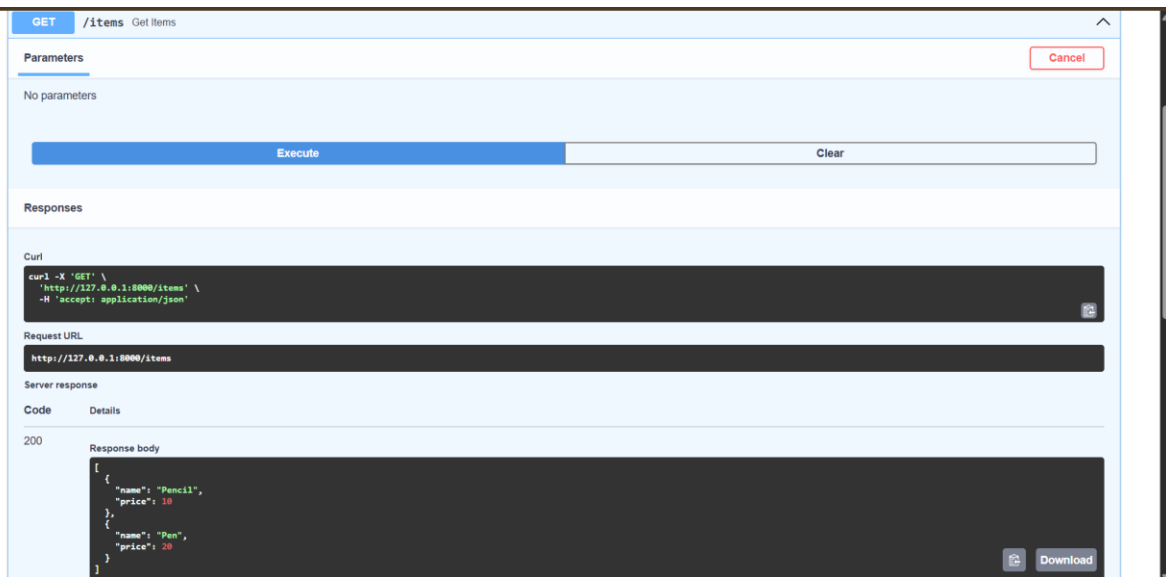
Response body

```
{
  "message": "Item updated",
  "item": {
    "name": "Pencil",
    "price": 10
  }
}
```

Download

Response headers

```
content-length: 62
content-type: application/json
date: Tue, 17 Mar 2020 15:50:10 GMT
server: unicorn
```

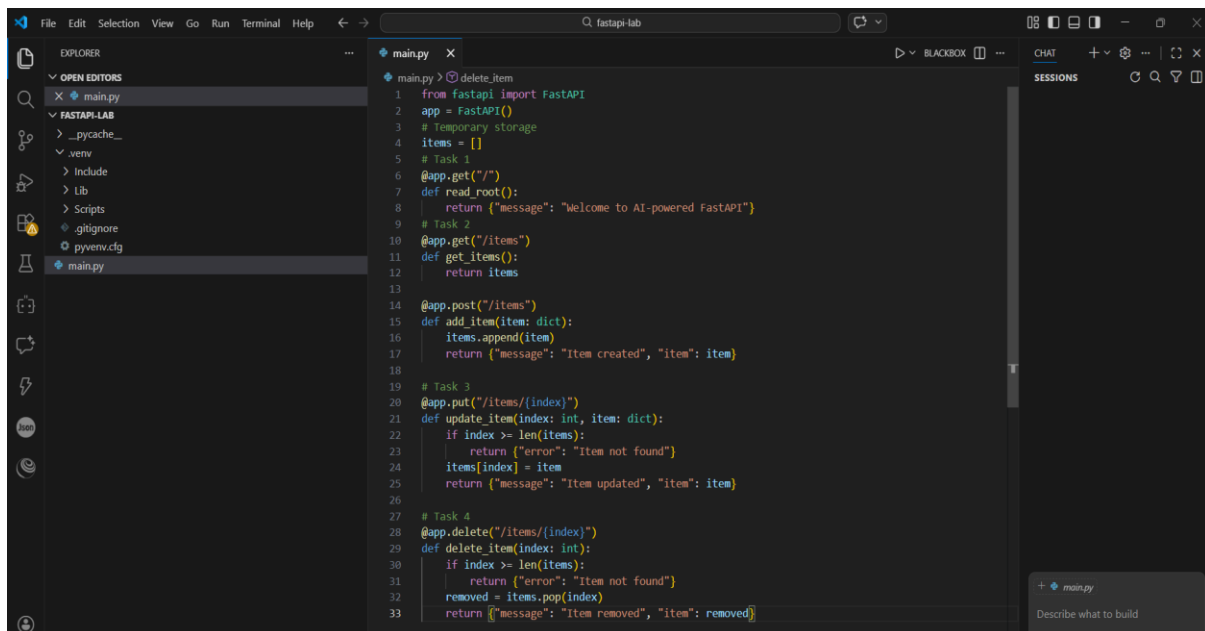


Explanation: PUT modifies an existing item in the list. It checks whether the given index exists.

Task Descriptor- 4: (Delete Item)

PROMPT: Create a DELETE endpoint to remove an item using its index.

INPUT:



OUTPUT:

```
PS C:\Users\sarik\Downloads\fastapi-lab> & C:\Users\sarik\Downloads\fastapi-lab\.venv\Scripts/Activate.ps1
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\Users\sarik\Downloads\fastapi-lab']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reload process [12756] using Sockload
INFO: Started server process [21644]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [21644]
INFO: Stopping reload process [12756]
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab>
```

POST /items Add item

Parameters

No parameters

Request body required application/json

Edit Value | Schema

```
{
  "name": "Pen",
  "price": 20
}
```

Execute Clear

Responses

GET /items Get items

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

Code Details

200

Response body

```
{
  "name": "Pencil",
  "price": 10
},
{
  "name": "Pen",
  "price": 20
},
{
  "name": "Bob"
}
```

DELETE

/items/{index} Delete Item

Cancel

Parameters

Name	Description
index required	
integer	
(path)	

Execute

Clear

Responses

Curl

Request URL

Server response

Code

Details

200

Response body

Response headers

Responses

Code

Description

Links

200

Successful Response

Media type

application/json

Controls Accept header

No links

GET

/items Get Items

Cancel

Parameters

No parameters

Execute

Clear

Responses

Curl

Request URL

Server response

Code

Details

200

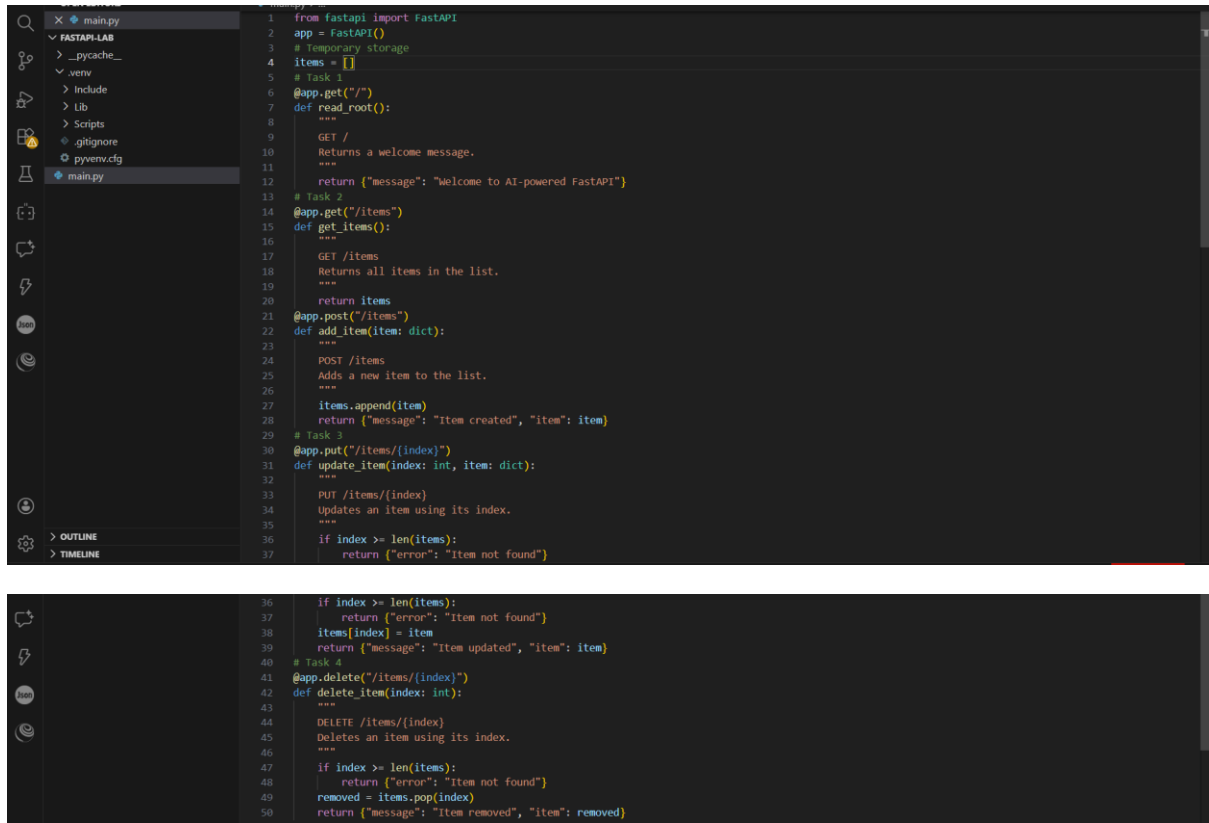
Response body

Explanation: DELETE removes an item from the list. It returns the deleted item as response.

Task Descriptor -5: (Add Auto-Generated Documentation)

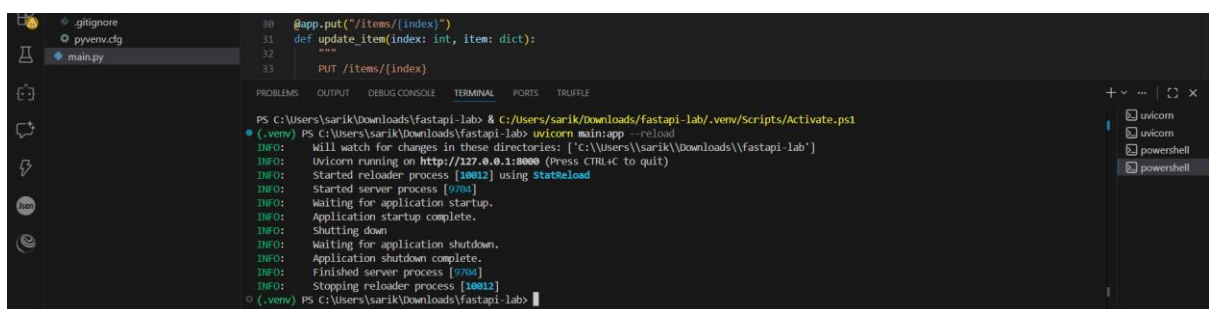
PROMPT: Add docstrings to endpoints for automatic Swagger documentation.

INPUT:



```
1 from fastapi import FastAPI
2 app = FastAPI()
3 # Temporary storage
4 items = []
5 # Task 1
6 @app.get("/")
7 def read_root():
8     """
9     GET /
10    Returns a welcome message.
11    """
12    return {"message": "Welcome to AI-powered FastAPI"}
13 # Task 2
14 @app.get("/items")
15 def get_items():
16     """
17     GET /items
18     Returns all items in the list.
19     """
20    return items
21 @app.post("/items")
22 def add_item(item: dict):
23     """
24     POST /items
25     Adds a new item to the list.
26     """
27    items.append(item)
28    return {"message": "Item created", "item": item}
29 # Task 3
30 @app.put("/items/{index}")
31 def update_item(index: int, item: dict):
32     """
33     PUT /items/{index}
34     Updates an item using its index.
35     """
36    if index >= len(items):
37        return {"error": "Item not found"}
38    items[index] = item
39    return {"message": "Item updated", "item": item}
40 # Task 4
41 @app.delete("/items/{index}")
42 def delete_item(index: int):
43     """
44     DELETE /items/{index}
45     Deletes an item using its index.
46     """
47    if index >= len(items):
48        return {"error": "Item not found"}
49    removed = items.pop(index)
50    return {"message": "Item removed", "item": removed}
```

OUTPUT:



```
PS C:\Users\sarik\Downloads\fastapi-lab> & C:\Users\sarik\Downloads\fastapi-lab\.venv\Scripts\Activate.ps1
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\Users\sarik\Downloads\fastapi-lab']
INFO: Uvicorn running on http://127.0.0.1:8000 (press CTRL+C to quit)
INFO: Started reload process [10012] using StatReload
INFO: Started server process [9706]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [9706]
INFO: Stopping reload process [10012]
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab>
```

default

GET	/	Read Root	⌵
GET	/items	Get Items	⌵
POST	/items	Add Item	⌵
PUT	/items/{index}	Update Item	⌵
DELETE	/items/{index}	Delete Item	⌵

Schemas

HTTPValidationError > Expand all object

ValidationError > Expand all object

default

GET / Read Root

Parameters Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/

Server response

Code	Details
200	Response body <pre>{ "message": "Welcome to AI-powered FastAPI" }</pre> Download

Response headers

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/

Server response

Code	Details
200	Response body <pre>{ "message": "Welcome to AI-powered FastAPI" }</pre> Download Response headers <pre>content-length: 43 content-type: application/json date: Tue, 17 Apr 2024 15:24:45 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header

Example Value | Schema

```
"string"
```

GET /items Get Items

Parameters

Cancel

No parameters

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/items' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

CodeDetails

200

Response body

```
[ ]
```

Download

Response headers

```
content-length: 2
content-type: application/json
date: Tue, 17 Mar 2025 15:25:59 GMT
```

POST /items Add Item

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "name": "Pen",
  "price": 20
}
```

Execute

Responses

CodeDescriptionLinks

GET /items Get Items

Parameters

Cancel

No parameters

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/items' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

CodeDetails

200

Response body

```
{
  "name": "Pen",
  "price": 20
},
{
  "name": "Pen",
  "price": 20
}

```

Download

PUT /items/{index} Update Item

Parameters

Cancel

Reset

Name	Description
index required	
integer	
(path)	0

Request body required

application/json

Edit Value

Schema

```
{  "name": "Pencil",  "price": 10}
```

Execute

GET /items Get Items

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \  'http://127.0.0.1:8000/items' \  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/items

Server response

Code	Details
200	Response body <pre>{ { "name": "Pencil", "price": 10 }, { "name": "Pen", "price": 20 } }</pre> Download

DELETE /items/{index} Delete Item

Parameters

Cancel

Name	Description
index required	
integer	
(path)	0

Execute

Clear

Responses

Curl

```
curl -X 'DELETE' \  'http://127.0.0.1:8000/items/0' \  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/items/0

Server response

Code	Details
------	---------

Postman interface showing a DELETE request to `http://127.0.0.1:8000/items/0`. The response is a 200 status code with a JSON body:

```
{
  "message": "Item removed",
  "item": {
    "name": "Pencil",
    "price": 10
  }
}
```

The response headers are:

```
content-length: 62
content-type: application/json
date: Tue, 17 Mar 2026 16:01:00 GMT
server: uvicorn
```

The response is categorized as a "Successful Response" with a media type of `application/json`.

Postman interface showing a GET request to `/items`. The response is a 200 status code with a JSON body:

```
{
  "name": "Pen",
  "price": 20
},
{
  "name": "Pen",
  "price": 20
},
{
  "name": "Pen",
  "price": 20
}
```

Explanation: FastAPI uses docstrings to generate API documentation. Swagger UI is available at `/docs`.